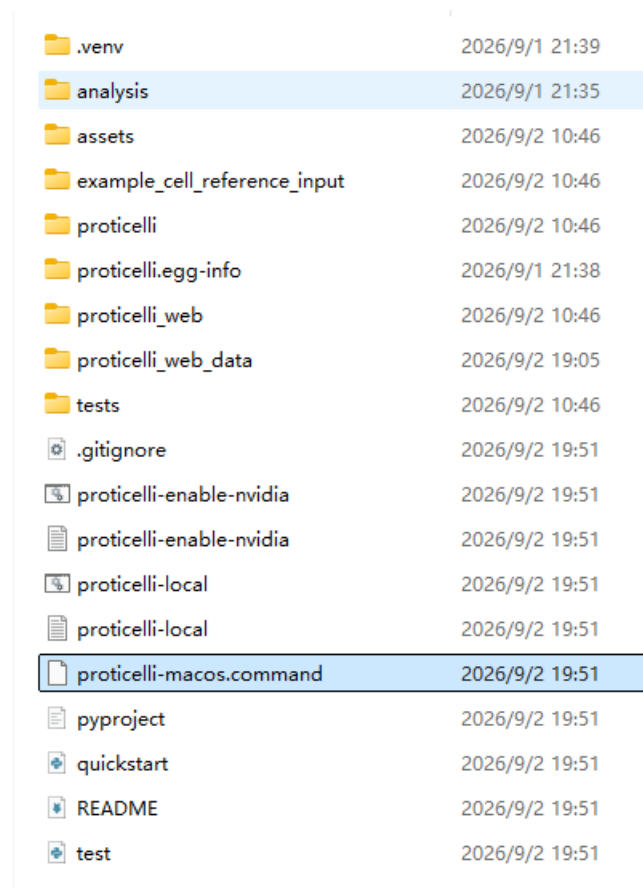


ProtiCelli Interactive Gallery

Local installation and quick-start handbook

The Gallery runs locally at `http://127.0.0.1:8000`. Reference images, workspaces, predictions, and exports remain on this computer by default.

1. Extract the installation package



Folder	.venv	2026/9/1 21:39
Folder	analysis	2026/9/1 21:35
Folder	assets	2026/9/2 10:46
Folder	example_cell_reference_input	2026/9/2 10:46
Folder	proticelli	2026/9/2 10:46
Folder	proticelli.egg-info	2026/9/1 21:38
Folder	proticelli_web	2026/9/2 10:46
Folder	proticelli_web_data	2026/9/2 19:05
Folder	tests	2026/9/2 10:46
File	.gitignore	2026/9/2 19:51
File	proticelli-enable-nvidia	2026/9/2 19:51
File	proticelli-enable-nvidia	2026/9/2 19:51
File	proticelli-local	2026/9/2 19:51
File	proticelli-local	2026/9/2 19:51
File	proticelli-macos.command	2026/9/2 19:51
File	pyproject	2026/9/2 19:51
File	quickstart	2026/9/2 19:51
File	README	2026/9/2 19:51
File	test	2026/9/2 19:51

Important: Extract the ZIP into a normal writable folder before launching. Do not run the application from inside the compressed ZIP.

2. Launch the Gallery for your operating system

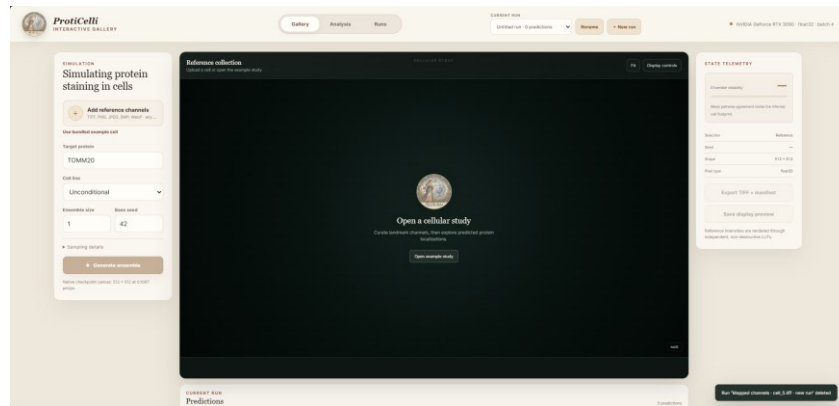
- WINDOWS:** Double-click `proticelli-local.bat`.
- MACOS:** Double-click `proticelli-macos.command`. If macOS blocks it, right-click the file and choose Open.
- LINUX:** Run `chmod +x proticelli-local.sh` once, then run `./proticelli-local.sh`. Use the NVIDIA helper only if the runtime indicator reports CPU-only PyTorch despite an NVIDIA GPU:
 - WINDOWS:** Run `proticelli-enable-nvidia.bat` once.
 - LINUX:** Run `chmod +x proticelli-enable-nvidia.sh`, then `./proticelli-enable-nvidia.sh` once.

On first launch, the installer creates `.venv`, installs dependencies, downloads and validates the checkpoint and VAE, then starts the Gallery. Keep the window open.

Later launches reuse everything; if setup is interrupted, run the same launcher again.

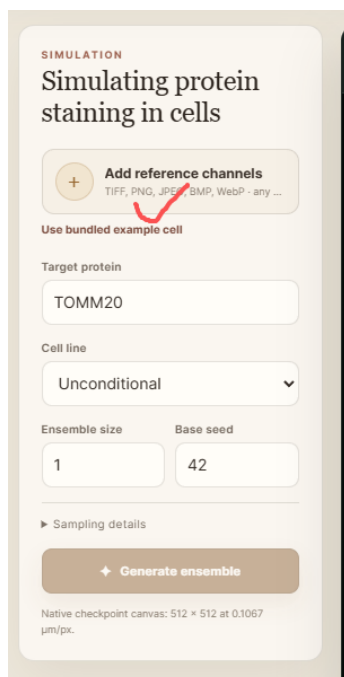
macOS: Apple Silicon uses Metal (MPS) automatically and does not use the NVIDIA helper.
After any GPU helper finishes, restart the Gallery.

3. Open the ProtiCelli Interactive Gallery



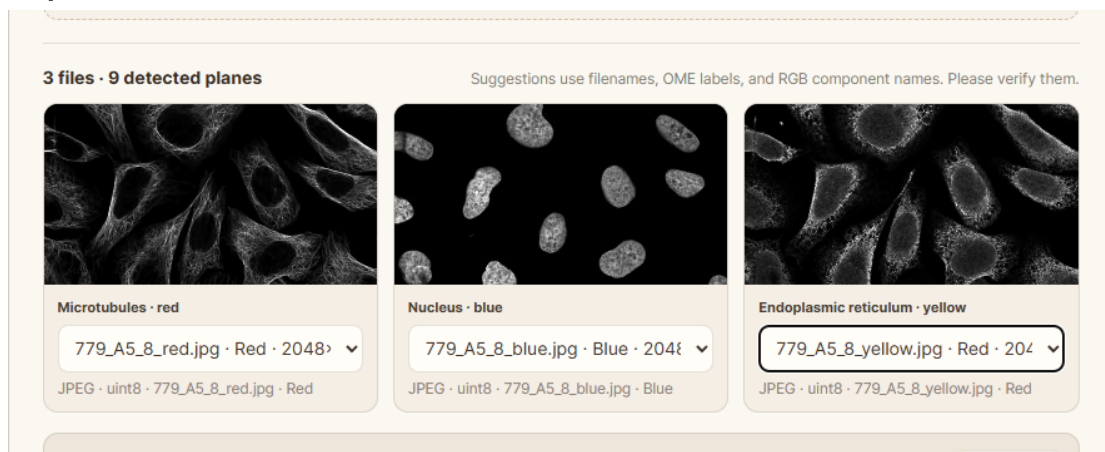
Browser: If the page does not open automatically, keep the launcher running and visit <http://127.0.0.1:8000>. Click the top-right runtime indicator to confirm the selected device.

4. Load reference images



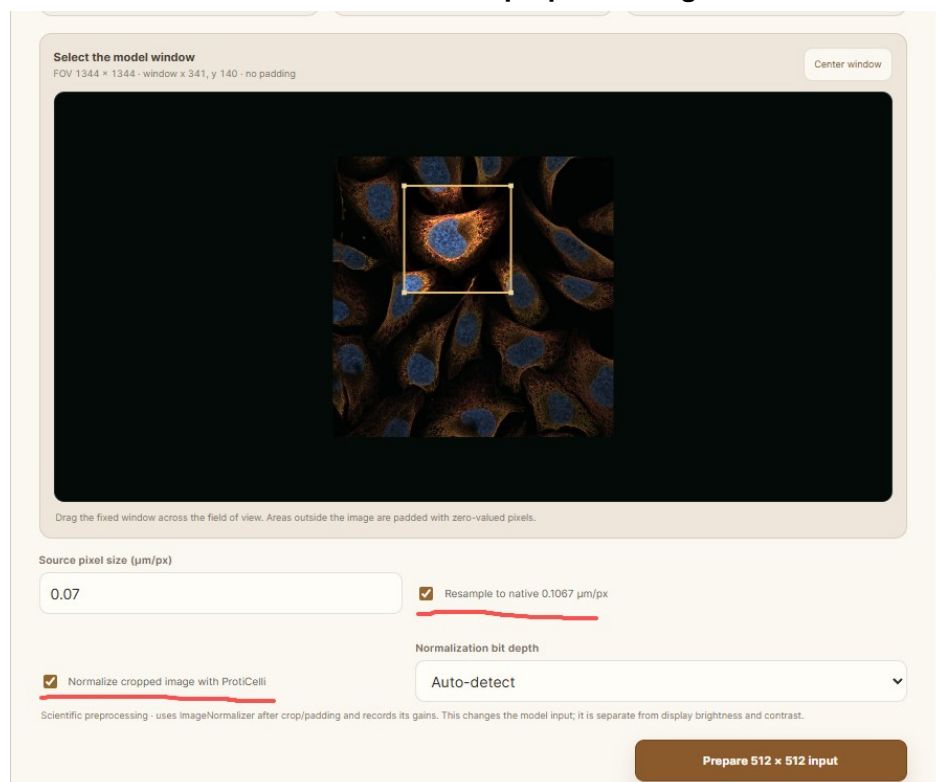
Supported input: Use one multichannel image or several single-channel files: TIFF/OME-TIFF, PNG, JPEG/JFIF, BMP, WebP, or static GIF. Prefer TIFF or PNG for quantitative intensities; JPEG is lossy.

Map the three reference channels



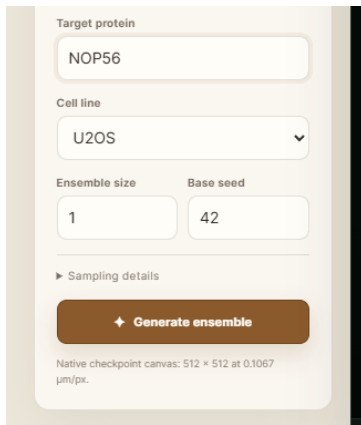
Channel mapping: Verify every automatic suggestion. Map microtubules (MT), nucleus, and endoplasmic reticulum (ER). Their default display colors are red, blue, and yellow; the predicted protein is green.

5. Select the 512 x 512 model window and preprocessing



Crop and normalize: Drag the fixed window over the field of view. Regions outside smaller images are zero-padded. Enter the true source pixel size before resampling. Optional ProtiCelli normalization changes the model input; brightness and contrast controls later are display-only.

6. Choose the simulation conditions



Target protein

NOP56

Cell line

U2OS

Ensemble size

1

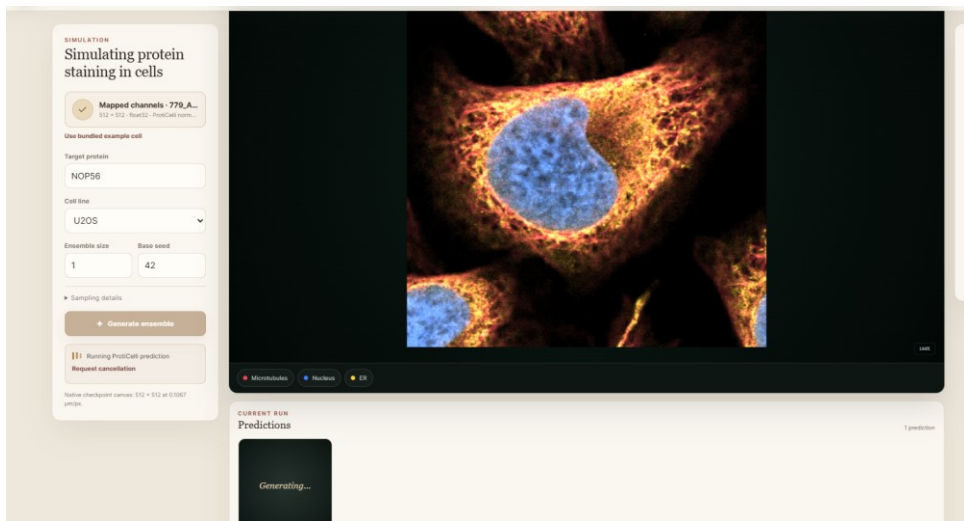
Base seed

42

► Sampling details

◆ Generate ensemble

Native checkpoint canvas: 512 × 512 at 0.1067 $\mu\text{m}/\text{px}$.



SIMULATION
Simulating protein staining in cells

Mapped channels - 779_A_...
512 × 512 - float32 - ProCell norm...

Use bundled example cell

Target protein

NOP56

Cell line

U2OS

Ensemble size

1

Base seed

42

► Sampling details

◆ Generate ensemble

Running ProCell prediction.
Request cancellation

Native checkpoint canvas: 512 × 512 at 0.1067 $\mu\text{m}/\text{px}$.

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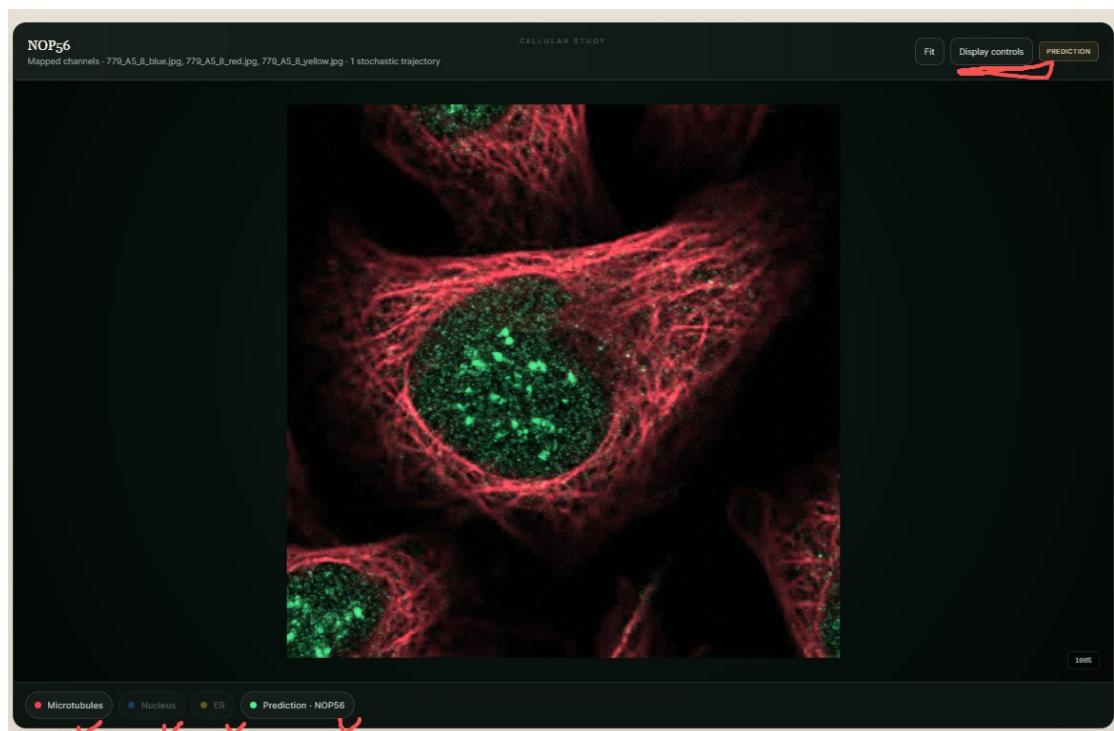
CURRENT RUN
Predictions

Generating...

1 prediction

Sampling: Choose a target protein and optionally a cell line; Unconditional is the default. Ensemble size defaults to 1, and the base seed makes repeated settings reproducible. Sampling details use float32 by default; float16 is a faster, lower-VRAM CUDA/ROCm option.

7. Inspect the generated protein staining



Display controls: Toggle channels, choose custom colors, pan or zoom, and adjust each channel's LUT independently. These adjustments never modify the raw float32 prediction.

8. Generate a reliability-aware ensemble

SIMULATION
Simulating protein staining in cells

✓ Mapped channels - 779_A...
512 × 512 - float32 - ProtiCell norm...

Use bundled example cell

Target protein
CS

Cell line
U2OS

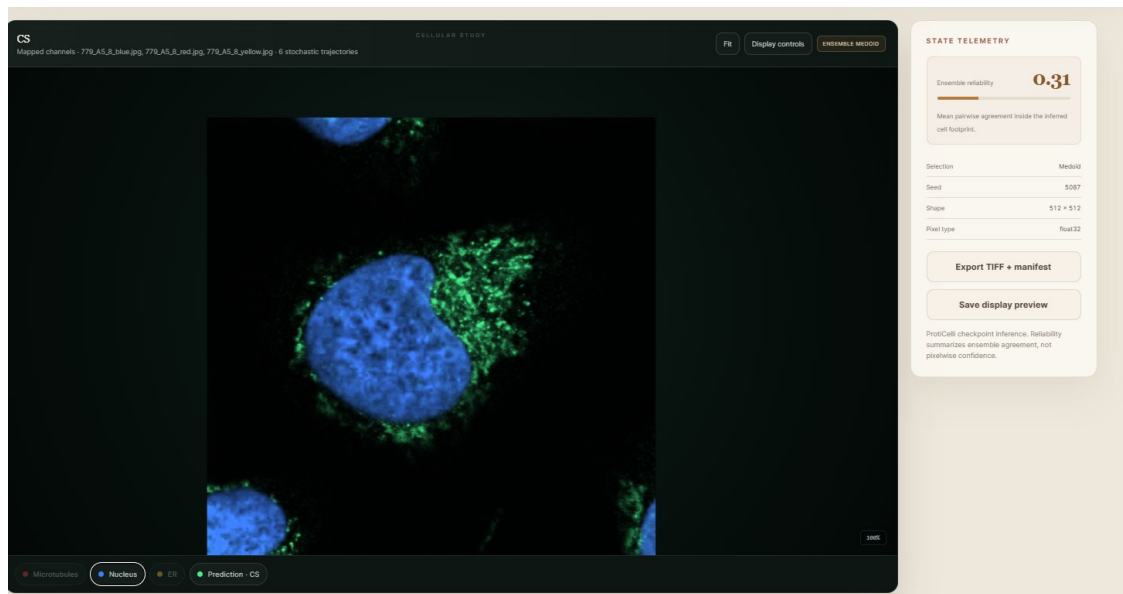
Ensemble size
6

Base seed
42

► Sampling details

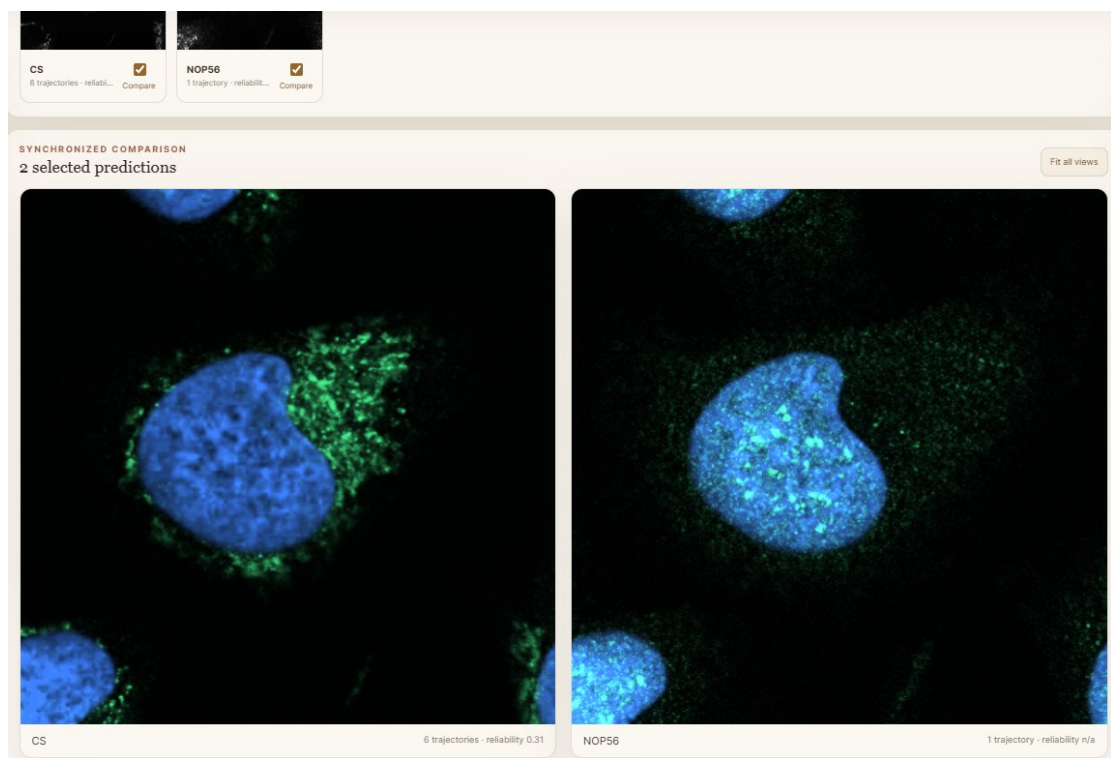
✦ Generate ensemble

Native checkpoint canvas: 512 × 512 at 0.1067 $\mu\text{m}/\text{px}$.



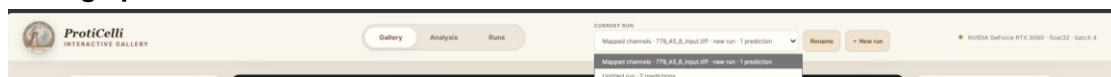
Reliability: Set Ensemble size to 2 or more. The Gallery shows stochastic alternatives, selects an ensemble medoid, and reports global ensemble agreement. This value is not pixel-level confidence. Larger ensembles take longer and use more disk space; the maximum is 50.

9. Compare predictions side by side



Comparison: Select two or more completed prediction cards. The comparison views share one pan and zoom transform, so localization patterns remain spatially aligned.

10. Manage persistent runs



RUN	REFERENCE	CONDITIONS	PREDICTIONS	ACTIVE	UPDATED
Mapped channels - 778_A5L8_input.tif - new run	Mapped channels - 778_A5L8_input.tif	Unconditional	1	---	2026/6/2 20:13:16
Untitled run	Mapped channels - 778_A5L8_black.jpg, 778_A5L8_red.jpg, 778_A5L8_green.jpg	UDOS	2	---	2026/6/2 20:11:05

Runs: A run keeps one reference-cell context and all predictions made from it; the cell line is selected per prediction. After the first prediction, the reference is locked. Runs can be renamed, duplicated, or deleted. Deleting removes its prediction artifacts but retains shared reference inputs, and active inference must be stopped first.

11. Export results

Scientific export: Export TIFF + manifest saves raw samples and reproducibility metadata, including the input hash, preprocessing, conditioning, seeds, precision, and ensemble statistics. Save display preview exports the current colors and LUT as PNG.

Quick troubleshooting

GPU not used: Click the runtime indicator or run `proticelli-web --diagnose`. On Windows/Linux with NVIDIA, run the matching `enable-nvidia` helper once and restart the Gallery.

Slow first prediction: The first inference loads and warms the model. Later predictions reuse the resident model. Cancellation is cooperative and completes after the current denoising step returns.

Local data: Workspaces and generated files are stored in `proticelli_web_data`. Back up that folder to preserve runs when moving the application to another computer.